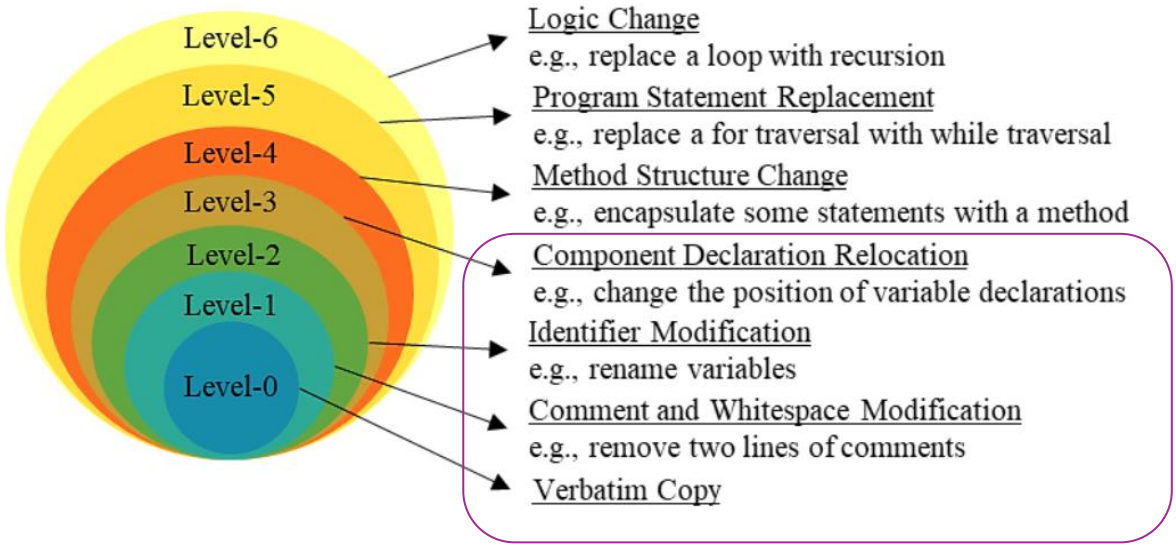
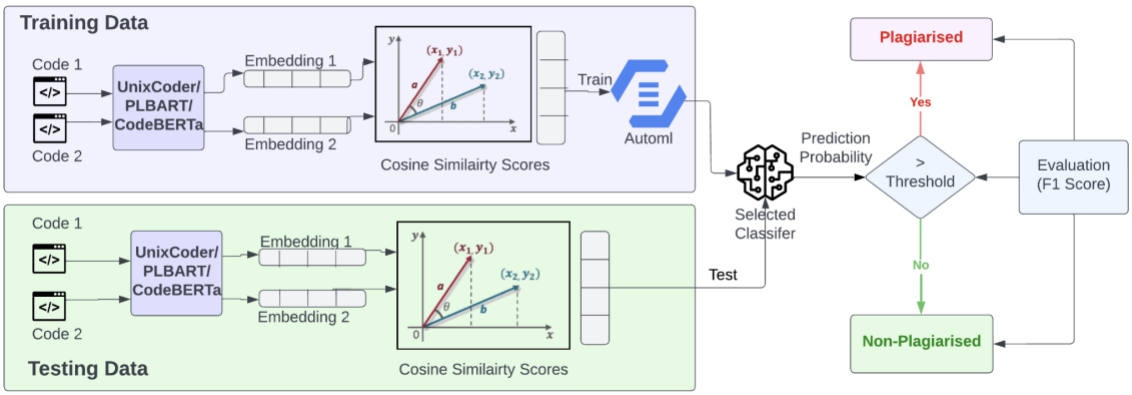
## Level of code plagiarsim:



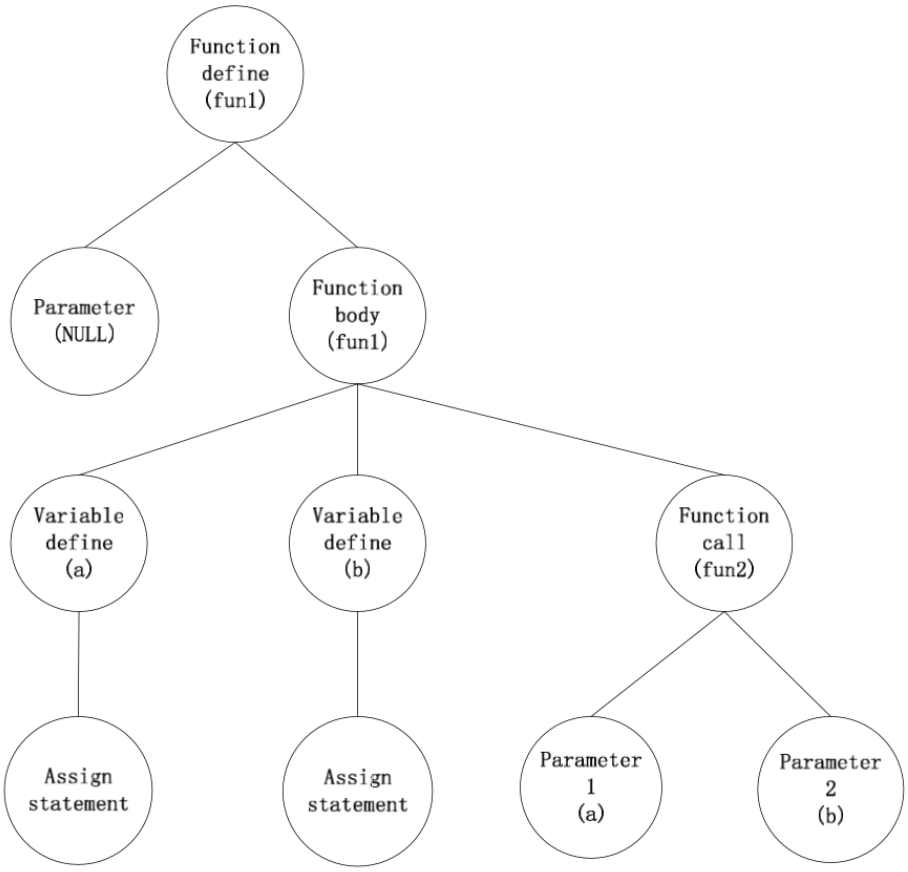

## Code plagiarsim pipeline:



## AST Example:

**Algorithm 1** Source code

```
void fun1( )
{
    int a = 2;
    int b = 5;
    fun2(a, b);
}
```



## *Embedding Generation from AST:*

```
//main.c

main()

{

inta,b;

}
```

**Fig 4: Source code of main.c**

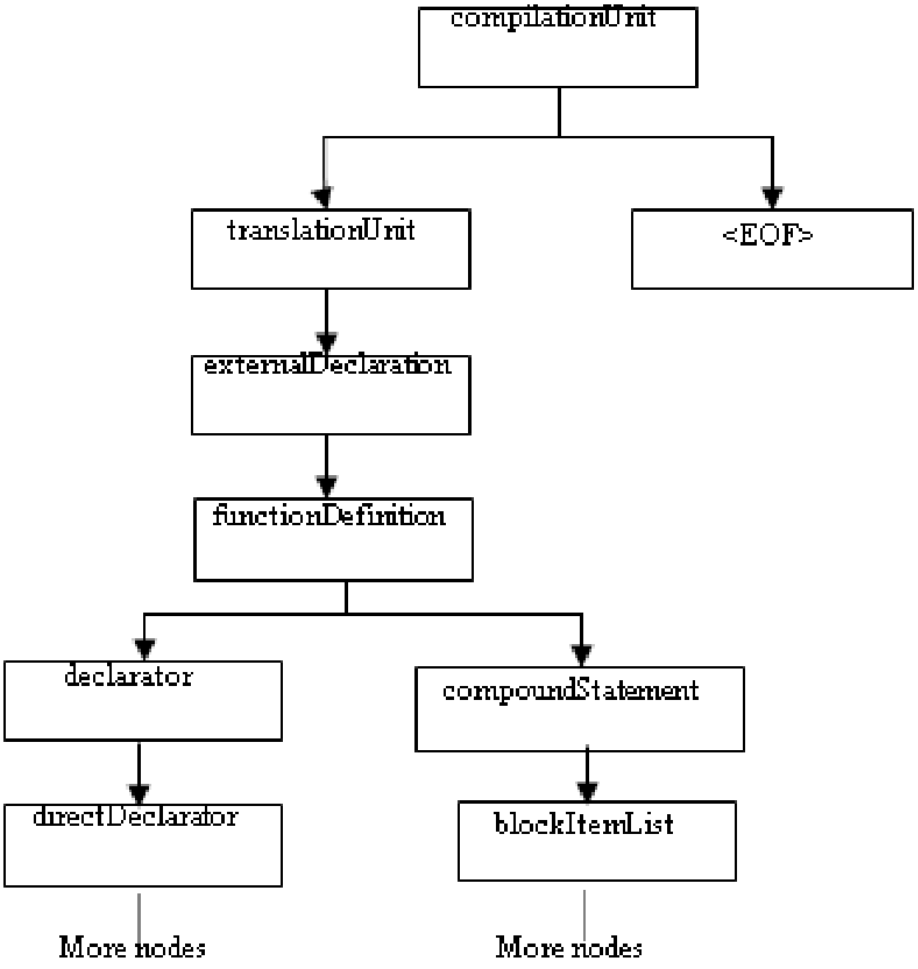



**Fig 5: AST generated from main.c**

**Token sequence:**

compilationUnit-translationUnit-externalDeclaration-functionDefinition-declarator-directDeclarator-directDeclarator-compoundStatement–blockItemList-blockItem-declaration-declarationSpecifiers-declarationSpecifier-typeSpecifier-initDeclaratorList-initDeclaratorList-initDeclarator-declarator-directDeclarator-initDeclarator-declarator-directDeclarator-EOF

## ***Some important points:***

- **Target Audience**: Designed for university teachers and assistants.
- **Focus on Large Projects**: Effective for semester and year-long coursework.
- **Intricate Logic**: Larger projects feature complex logic and multiple components.
- **Variability**: Small assignments have less code, leading to similar solutions.
- **False Positives**: More potential for false positives in smaller assignments.
- **Lack of Depth**: Smaller tasks don't showcase unique approaches or deep understanding.
- **Balancing Sensitivity and Specificity**: Crucial to avoid false positives in detection.

## ***Why not smaller projects?:***

- **Simplicity of Task**: Simple functions (e.g., adding two numbers) have limited implementation styles, leading to high similarity without indicating plagiarism.
- **Common Patterns**: Many students use similar variable names and structures, resulting in false positives due to the nature of the task.
- **Lack of Context**: Smaller assignments often lack the complexity needed for meaningful originality assessment.
- **Educational Value**: These tasks are typically practice-focused; plagiarism detection may not yield valuable insights for educators.
- **Future Considerations**: Consider developing methods for small assignments later, possibly by examining broader patterns across multiple submissions.